

Professional Exposure to Pesticides and Parkinson Disease

Alexis Elbaz, MD, PhD,^{1,2,3} Jacqueline Clavel, MD, PhD,^{4,5} Paul J. Rathouz, PhD,⁶ Frédéric Moisan, MSc,^{1,2} Jean-Philippe Galanaud, MD,^{1,2} Bernard Delemotte, MD,⁷ Annick Alpérovitch, MSc, MD,^{1,2} and Christophe Tzourio, MD, PhD^{1,2}

Objective: We studied the relation between Parkinson disease (PD) and professional exposure to pesticides in a community-based case-control study conducted in a population characterized by a high prevalence of exposure. Our objective was to investigate the role of specific pesticide families and to perform dose-effect analyses.

Methods: PD cases ($n = 224$) from the Mutualité Sociale Agricole (France) were matched to 557 controls free of PD affiliated with the same health insurance. Pesticide exposure was assessed using a 2-phase procedure, including a case-by-case expert evaluation. Analyses of the relation between PD and professional exposure to pesticides were first performed overall and by broad category (insecticides, fungicides, herbicides). Analyses of 29 pesticide families defined based on a chemical classification were restricted to men. Multiple imputation was used to impute missing values of pesticide families. Data were analyzed using conditional logistic regression, both using a complete-case and an imputed dataset.

Results: We found a positive association between PD and overall professional pesticide use (odds ratio [OR] = 1.8, 95% confidence interval [CI] = 1.1–3.1), with a dose-effect relation for the number of years of use ($p = 0.01$). In men, insecticides were associated with PD (OR = 2.2, 95% CI = 1.1–4.3), in particular organochlorine insecticides (OR = 2.4, 95% CI = 1.2–5.0). These associations were stronger in men with older onset PD than in those with younger onset PD, and were characterized by a dose-effect relation in the former group.

Interpretation: Our results support an association between PD and professional pesticide exposure, and show that some pesticides (ie, organochlorine insecticides) may be more particularly involved.

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The etiology of Parkinson disease (PD) is likely to be multifactorial; although genetic susceptibility plays a role, as demonstrated by familial aggregation and association studies,^{1,2} twin studies and segregation analyses suggest that environmental factors are also involved.^{3,4}

Studies of the relation between pesticides and PD were triggered by the description of parkinsonism in drug users exposed to 1-methyl-4-phenyl-1,2,3,6 tetrahydropyridine (MPTP),⁵ which also causes degeneration of dopaminergic neurons in animals.⁶ To be active, MPTP is metabolized into 1-methyl-4-phenylpyridinium, which has a chemical structure similar to that of the herbicide paraquat.

Case-control and cohort studies have shown an association between pesticides and PD, and specific pesticides have been shown to be toxic for dopaminergic neurons.⁷ Although accumulating data support this as-

sociation, the majority of epidemiologic studies have neither implicated specific pesticides nor assessed dose-effect relations.⁸

We studied the relation between PD and pesticides in a community-based case-control study conducted in a population characterized by a high prevalence of exposure. Our objective was to study in detail the relation between PD and professional pesticide exposure, by investigating pesticide families and performing dose-effect analyses.

Materials and Methods

This study was conducted among affiliates to Mutualité Sociale Agricole (MSA), the French health insurance for workers in agriculture and related occupations while active or retired (farm owners and workers; workers in silos, agricultural cooperatives, seed shops; tertiary sector professionals).^{9,10} At

From ¹INSERM, U708, Paris, France; ²UPMC, Univ Paris 06, Paris, France; ³Département santé-travail, Institut de Veille Sanitaire, Saint-Maurice, France; ⁴INSERM, U754, Villejuif, France; ⁵Univ Paris 11, Villejuif, France; ⁶Department of Health Studies, University of Chicago, Chicago, IL; and ⁷Mutualité Sociale Agricole, Bagnolet, France.

Additional Supporting Information may be found in the online version of this article.

Potential conflict of interest: Nothing to report.

Address correspondence to Dr Elbaz, Inserm, U708, Hôpital de la Salpêtrière, 47 boulevard de l'Hôpital, 75651 Paris Cedex 13, France. E-mail: alexis.elbaz@upmc.fr

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the time of the study, MSA counted approximately 4 million members and 82 offices distributed across France, according to an administrative division (French “départements”); 62 offices agreed to participate. The research protocol was approved by the ethics committee of Hôpital du Kremlin-Bicêtre, and all subjects signed an informed consent.

Participants

In France, free healthcare is available for PD. Eligible patients (18–75 years old) applied for free healthcare for PD to MSA (February 1998–August 1999). Patients with free healthcare for dementia were not eligible.

Patients who agreed to participate were examined by a movement disorders neurologist. When not possible, patients’ neurologists were contacted. Parkinsonism was defined as the presence of ≥ 2 cardinal signs (rest tremor, bradykinesia, rigidity, impaired postural reflexes). PD was defined by parkinsonism after exclusion of prominent or early signs of more extensive nervous system involvement and drug-induced parkinsonism.¹¹

Eligible controls were affiliates who requested reimbursement for health-related expenses (February 1998–February 2000). A maximum of 3 controls were matched to each case (on age ± 2 years, sex, affiliation office). Subjects with free healthcare for dementia or PD were not eligible.

Data Collection

A physician interviewed the participants (to obtain information on demographic data, smoking, well water drinking, and family history)¹⁰ examined the participants to confirm the presence (patients) or absence (controls) of parkinsonism, and administered the Mini Mental State Examination (MMSE).

Professional History and Pesticide Exposure

Pesticide exposure was assessed using a 2-phase procedure.^{12,13} First, occupational history was obtained through self-questionnaires. For each job, participants declared if they had personally sprayed pesticides. Information on pesticide use for gardening was also obtained. Questionnaires were reviewed, blinded to case-control status, to classify participants as never users of pesticides, users for gardening only, or professional users. We interviewed never users and users for gardening over the phone ($n = 30$) to confirm that none of them had been professional users.

Second, professional users were interviewed at home by an MSA occupational health physician to reconstruct the history of pesticide use. Participants were asked to list each farm where they had worked (start/end years). They were asked to describe each farm in terms of land size, crops (size), and animal breeding (number), and whether they had personally sprayed pesticides for each of them. For each crop/animal for which pesticides were used, detailed information was obtained: pesticides, frequency (days/year), duration (hours/year), spraying method (portable device, tractor), and start/end years. To use as many sources of information as possible, interviewers visited the farm, discussed technical issues, looked for old pesticide containers and packages, and reviewed bills and farming calendars. Farm interviews usually lasted half a day or more. Ten subjects were reinterviewed 6

months later; agreement was excellent for all variables used in analysis.

Home interview questionnaires were reviewed by 2 epidemiologists, 2 occupational health physicians, and an agronomist blinded to disease status. Special attention was given to: pesticide availability at the reported date of use, likelihood of use for the target crop/animal, and likelihood of duration/frequency of use given the land size or number of animals. In case of implausibility, the occupational health physician recontacted the participant.

Coding of Pesticides and Measures of Exposure

Using these questionnaires, we constructed an “exposure history” dataset consisting of multiple observations per subject, each corresponding to an instance of pesticide use. Each observation included several variables: pesticide, crop/animal, spraying method, frequency (days/year), duration (hours/year), and start/end years.

Pesticides were coded using a pesticide dictionary and were grouped into 1) 50 families based on chemical similarities (<http://www.alanwood.net/pesticides>): 12 insecticides (eg, organophosphorus, organochlorines), 21 herbicides (eg, dinitrophenol, nitrile), and 17 fungicides (eg, dithiocarbamate, triazole); and 2) 3 pesticide categories: insecticides, fungicides, and herbicides. Four generic “unknown” codes were used when a subject did not remember the specific pesticide used (unknown insecticide, fungicide, herbicide, and pesticide). Fifty-three indicators were computed (for 50 families and 3 categories). When generic “unknown” codes were used, all corresponding family indicators were coded as missing, as any of them could have been used (eg, for an “unknown insecticide” code, the “insecticide” indicator was coded “Yes,” the 12 indicators for insecticide families were coded “Missing,” and the indicators for “fungicide,” “herbicide,” and 38 families of fungicides and herbicides were coded “No”).

Finally, an “exposure summary” dataset with 1 line/subject was obtained, including 53 yes/no indicators for pesticide use; for each indicator, we computed variables assessing exposure intensity (cumulative lifetime number of years/days/hours of exposure; cumulative hectares $\times$ years/days/hours of exposure).¹⁴ Only exposures occurring before PD onset in cases and index age (age at onset in matched case) in controls were included.

Statistical Methods

We first conducted complete-case analyses comparing the frequency of professional use of pesticides, insecticides, fungicides, herbicides, and pesticide families by disease status, using conditional logistic regression (proc PHREG, SAS v9.1, SAS Institute Inc., Cary, NC).¹⁵ Continuous variables were dichotomized using the median of their distribution in exposed subjects; trend tests were performed using a 3-level variable (0 for no exposure, and midpoints of intervals below and above the median among those exposed). An indicator variable for “gardening-only” pesticide use was included to exclude subjects exposed exclusively through gardening from the reference group. All analyses were adjusted for MMSE and cigarette smoking (pack-years).¹⁰ Because exposure patterns were different in men and women, all analyses were

sex-stratified. Because genetic susceptibility may depend on age at onset, we stratified analyses by median age at onset/index age. As 1 study found an association between pesticides and incident but not prevalent PD,¹⁶ we also stratified analyses by median disease duration.

In complete-case analyses, observations with missing information for any of the predictors are deleted from the analyses.¹⁷ This approach reduces the precision of parameter estimates, may introduce bias, and yields different sample sizes depending on the predictors included in the model; the problem is especially severe in matched case-control studies. Simple approaches (eg, missing indicators method) do not in general correct this bias, and require that missing status be independent of case-control status.¹⁸ Multiple imputation (MI) is more generally appropriate, and allows asymptotically unbiased estimation of exposure effects under the weaker assumption that information is missing at random conditional on measured variables, including case-control status.¹⁷

We used MI to impute missing exposure values in the pesticide exposure history dataset. MI replaces missing values with plausible values based on observed data for that variable, case-control status, and other covariates. Across families of pesticides, the strongest predictors of pesticide use were the type of crop/animal and the time period of use. For each pesticide family indicator, we therefore imputed missing values using a logistic regression model with the following predictors: crop/animal, time period, case-control status, sex, age, and MMSE.¹⁹ For each indicator, MI was done independently of that for other indicators, as joint imputation across 50 families is computationally intractable. We used SAS PROC MI to generate 10 imputed exposure history datasets; each of these was then reduced to an exposure summary dataset. Imputed exposure summary data sets were analyzed as complete data, and the 10 sets of results were pooled using SAS PROC MIANALYZE.¹⁵ Note that MI allows multivariate exposure analyses because each of the datasets analyzed is complete.

Results

Of 352 PD patients who requested free healthcare, 292 agreed to participate (83%); 184 were examined by the study neurologist, who diagnosed PD in 153. Of the remaining 108, 94 were classified as having PD based on neurologists' reports. These 247 patients were matched to 676 controls (acceptance rate, 75%). Figure 1 describes the exposure assessment procedure. We were able to obtain detailed exposure data for 224 cases and 557 matched controls.

Cases had worked in farms more often than controls (Table 1). In analyses stratified by gender, the association between farming and PD was similar in men and women (interaction, $p = 0.36$). Cases had been professional pesticide users more often than controls; there was no significant difference in odds ratios (ORs) between men (OR = 1.8, 95% confidence interval [CI] = 0.9–3.8) and women (OR = 2.4, 95% CI = 1.1–5.6; interaction, $p = 0.54$). In men, ORs increased with the number of years of professional use ($p = 0.05$); the relation was stronger for male cases with on-

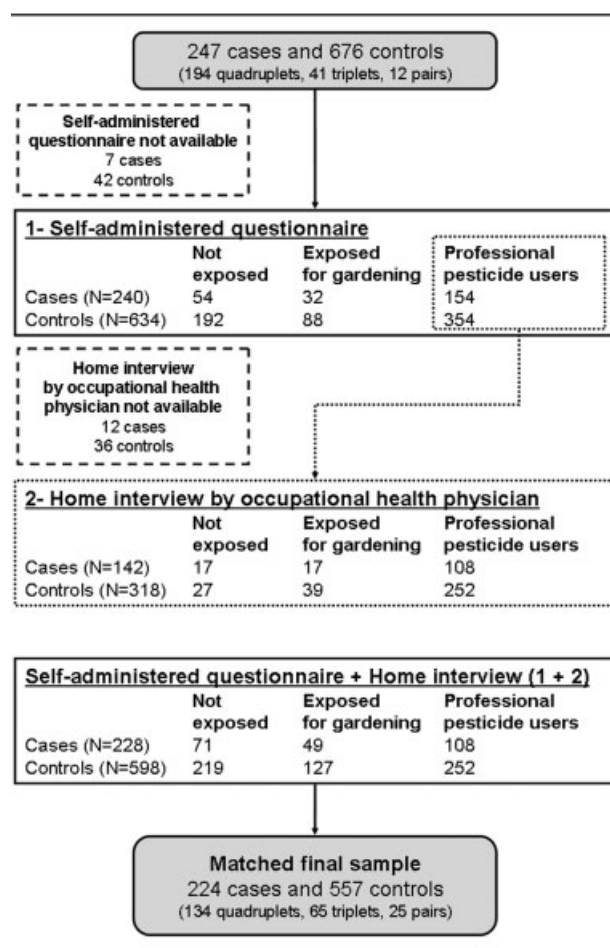


Fig 1. Flow chart for the assessment of exposure to pesticides.

set at >65 years (OR = 4.5, 95% CI = 1.0–20.0) than in those with younger onset (OR = 1.1, 95% CI = 0.5–2.7; interaction, $p = 0.05$).

Professionally exposed subjects generated a pesticide exposure history dataset with 4,322 observations; the median (interquartile range) number of observations/subject was 11 (13) in cases and 10 (13) in controls, was higher in men than in women, increased with MMSE, and was lower in older onset than in younger onset cases. The proportion of generic unknown codes per subject (unknown insecticide, fungicide, herbicide, or pesticide) was not significantly higher in cases (19%) than in controls (17%, $p = 0.24$); it increased with age and lower MMSE, and was higher in women than in men, both in cases and controls (interaction p values ranging from 0.35 to 0.46).

Table 2 shows ORs for exposure to 3 categories of pesticides (insecticides, fungicides, and herbicides); frequencies are shown in Appendix 1. Separate models were fitted for each category, and strata were defined by sex and age at onset. The proportion of missing values did not exceed 5% in any model. Frequency of

Table 1. General Characteristics of PD Patients and Controls

General Characteristics	Controls n = 557	Cases n = 224	OR (95% CI) ^a	p ^a
Median age at onset of PD, yr (IQ range)	—	65 (11)	—	—
Median age at diagnosis of PD, yr (IQ range)	—	66 (10)	—	—
Median age at the time of the study, yr (IQ range)	68 (9)	69 (10)	—	—
Median lag time between diagnosis and study, yr (IQ range)	—	1.5 (2.3)	—	—
Male sex, % (No.)	52 (291)	53 (118)	—	—
Median MMSE (IQ range)	28 (3)	27 (5)	—	<0.001
Cigarette smoking, % (No.) ^b				
Never cigarette smoking	69 (380)	77 (172)	1.0 (ref.)	—
Ever cigarette smoking	31 (170)	23 (50)	0.5 (0.4–0.8)	0.003
Ever cigarette smoking, pack-years ≤17	16 (85)	15 (32)	0.6 (0.4–1.1)	0.06
Ever cigarette smoking, pack-years >17	15 (85)	8 (18)	0.4 (0.2–0.7)	0.002
Farming, % (No.) ^c				
Never farming	29 (161)	18 (40)	1.0 (ref.)	—
Ever farming	71 (396)	82 (184)	1.9 (1.3–2.9)	0.003
Farming: No. of years ≤median (41 yr) ^d	36 (200)	33 (75)	1.4 (0.9–2.2)	0.15
Farming: No. of years >median (41 yr) ^d	35 (196)	49 (109)	2.7 (1.7–4.3)	<0.001
Pesticide exposure, % (No.) ^{e,f}				
No pesticide exposure	38 (211)	31 (69)	1.0 (ref.)	—
Exposure for gardening	22 (121)	21 (48)	1.4 (0.9–2.3)	0.18
Professional exposure	40 (225)	48 (107)	1.8 (1.1–3.1)	0.02
Professional exposure: No. of years ≤median (38 yr) ^d	21 (116)	24 (54)	1.7 (1.0–2.9)	0.05
Professional exposure: No. of years >median (38 yr) ^d	19 (109)	24 (53)	2.1 (1.1–3.9)	0.02
Ever well-water drinking, % (No.) ^g	64 (353)	65 (144)	1.0 (0.7–1.5)	0.83
Family history of PD (parents, siblings), % (No.) ^g	5 (27)	9 (20)	2.1 (1.1–3.9)	0.02

^aWe used conditional logistic regression for matched sets with a varying number of controls to compute ORs (95% CI). MMSE was categorized according to the quartiles of its pooled distribution to perform a trend test.

^bValues were missing for 7 controls and 2 cases. Trend test for pack-years, $p = 0.0008$.

^cFarming included farm owners, farm workers, and farm helpers. ORs increased with the number of years of farming (trend, $p < 0.001$).

^dORs were computed for each exposure category compared to the reference group (persons who were not farmers or who never used pesticides).

^eAnalyses for pesticides are based on the comparison of exposure status in 224 cases and 557 controls, because it was known for all cases and controls included in this paper.

^fAfter adjustment for MMSE level and cigarette smoking (pack-years), the ORs were: professional pesticide exposure, OR = 1.7 (95% CI = 1.0–2.9); professional exposure ≤38 years, OR = 1.6 (95% CI = 0.9–2.8); professional exposure >38 years, OR = 2.0 (95% CI = 1.0–3.5). ORs increased with the number of years of professional exposure (trend, $p = 0.01$).

^gValues were missing for 2 controls and 2 cases for both variables.

PD = Parkinson disease; OR = odds ratio; CI = confidence interval; IQ = interquartile; MMSE = Mini Mental State Examination; ref. = reference.

exposure to each of 3 categories was lower in women than in men. Male cases had used insecticides more often than controls; this relation was stronger for older onset PD, and was characterized by a dose-effect relation in this group. The relation between herbicides or fungicides (yes/no) and PD was not significant in men. There was a significant dose-effect trend for fungicides

in men with older onset PD. Among women, cases had used fungicides more frequently than controls. The number of exposed women was insufficient to perform more detailed analyses.

When insecticides and fungicides were included in the same model for men with older onset PD, insecticides (OR = 4.4, 95% CI = 1.0–20.4) but not fun-

Table 2. PD and Professional Exposure to 3 Broad Categories of Pesticides: Insecticides, Fungicides, and Herbicides

Professional Pesticide Exposure, % (No.)	Insecticides			Fungicides			Herbicides		
	OR (95% CI)	<i>p</i>	<i>p</i> ^a	OR (95% CI)	<i>p</i>	<i>p</i> ^a	OR (95% CI)	<i>p</i>	<i>p</i> ^a
Men									
No	1.0 (ref.)	—		1.0 (ref.)	—		1.0 (ref.)	—	
Yes	2.2 (1.1–4.3)	0.03		1.5 (0.8–3.0)	0.22		1.4 (0.7–2.6)	0.35	
Women									
No	1.0 (ref.)	—		1.0 (ref.)	—		1.0 (ref.)	—	
Yes	1.4 (0.5–3.8)	0.49		3.5 (1.2–10.3)	0.02		1.2 (0.4–3.8)	0.72	
Men, age at onset ≤median (65 yr)									
No	1.0 (ref.)	—		1.0 (ref.)	—		1.0 (ref.)	—	
Yes	1.2 (0.5–2.8)	0.67		1.2 (0.5–2.7)	0.72		0.9 (0.4–2.1)	0.81	
No. of cumulative hours <median	1.5 (0.6–3.9)	0.39		1.5 (0.6–3.8)	0.35		0.8 (0.3–2.0)	0.59	
No. of cumulative hours ≥median	0.9 (0.3–2.3)	0.77	0.44	0.7 (0.2–2.0)	0.49	0.17	0.9 (0.3–2.2)	0.75	0.95
Men, age at onset > median (65 yr)									
No	1.0 (ref.)	—		1.0 (ref.)	—		1.0 (ref.)	—	
Yes	4.9 (1.4–17.3)	0.01		2.7 (0.8–8.6)	0.10		2.1 (0.8–5.7)	0.13	
No. of cumulative hours <median	4.4 (1.2–16.8)	0.03		2.0 (0.6–7.0)	0.28		2.0 (0.7–6.0)	0.21	
No. of cumulative hours ≥median	5.4 (1.5–20.0)	0.01	0.04	4.8 (1.2–19.3)	0.03	0.02	2.2 (0.8–6.4)	0.14	0.24

ORs are based on separate conditional logistic regression models for matched sets comparing subjects exposed to any of the 3 broad pesticide categories to subjects who did not report that category of pesticide use (reference group); corresponding exposure frequencies and medians of the distributions of the cumulative lifetime number of hours of exposure in cases and controls are shown in Appendix 1. Subjects exposed to any pesticide for gardening exclusively are included in the analyses, but are treated as their own exposure group. All analyses are adjusted for Mini Mental State Examination and cigarette smoking (in pack-years).

^a*p*-value for trend (no exposure, <median, ≥median).

PD = Parkinson disease; OR = odds ratio; CI = confidence interval; ref. = reference.

gicides (OR = 1.2, 95% CI = 0.2–5.5) remained associated with PD. When the relation between PD and professional pesticide exposure was adjusted for insecticide use, the OR for professional pesticide exposure was 0.7 (95% CI = 0.1–8.7); adjustment for fungicides yielded an OR of 2.6 (95% CI = 0.3–25.6) for professional pesticide exposure. These analyses suggest that the association between PD and professional pesticide use among men with older onset PD was mainly explained by insecticide use rather than fungicide use; similar results were found among men overall (data not shown).

Appendix 2 shows frequencies in men of 29 pesticide families present in more than 5% of controls. Among controls, organochlorines, dithiocarbamates, and triazines were the most frequently reported insecticides, fungicides, and herbicides respectively. The mean percentage of missing values was: insecticides, 23%; herbicides, 20%; fungicides, 13%. When comparing controls matched to older and younger onset cases, frequencies of pesticide families were influenced

by their date of marketing. The mean cumulative lifetime number of years of exposure of all families was similar in controls matched to older onset cases (13 years) and in controls matched to younger onset cases (12 years), despite having a longer period of potential pesticide exposure in the older group. The mean cumulative lifetime number of hours of exposure of all families was higher in controls matched to younger onset cases (88 hours) than in those matched to older onset cases (52 hours).

Table 3 shows ORs for pesticide families among men overall and among those with older onset PD using a complete-case analysis; Appendix 3 shows the corresponding dose-effect analyses. Male cases had used organochlorine insecticides and amide or dithiocarbamate fungicides more often than controls; a dose-effect relation was present for nitrile herbicides. No association with younger onset PD was present for any family (data not shown). Older onset cases had used organochlorine insecticides and nitrile or phenoxy herbicides more often than controls; dose-effect relations were significant for

Table 3. PD and Exposure to 29 Families of Pesticides in Men: Complete-Case Analysis and Multiple Imputation of Missing Values

Exposure	Complete-Case Analysis		Multiple Imputation	
	Men, OR (95% CI)	Men, AO>65 yr, OR (95% CI)	Men, OR (95% CI)	Men, AO>65 yr, OR (95% CI)
Insecticides				
Carbamate	1.6 (0.7–3.7)	1.2 (0.3–4.7)	1.3 (0.6–2.7)	1.5 (0.4–5.0)
Organochlorine	2.4 (1.2–5.0) ^a	4.2 (1.5–11.9) ^a	1.9 (1.1–3.5) ^a	3.0 (1.2–7.9) ^a
Organophosphorus	1.8 (0.9–3.7)	2.9 (0.9–9.1)	1.3 (0.7–2.3)	1.6 (0.7–3.7)
Pyrethroid	1.4 (0.7–2.8)	1.8 (0.6–5.1)	1.0 (0.6–1.7)	1.1 (0.5–2.6)
Arsenic	1.5 (0.6–3.7)	1.6 (0.5–5.1)	1.6 (0.8–3.3)	2.7 (1.0–7.3) ^a
Fungicides				
Amide	3.1 (1.2–8.3) ^a	4.7 (0.9–23.0)	2.3 (0.9–5.9)	4.0 (0.9–19.1)
Aromatic	0.6 (0.2–2.3)	Not estimable	0.5 (0.1–1.8)	Not estimable
Carbamate	1.3 (0.6–2.6)	1.2 (0.3–5.4)	1.0 (0.5–1.8)	0.9 (0.3–2.8)
Dicarboximide	1.2 (0.4–3.0)	0.8 (0.2–3.4)	0.9 (0.4–2.0)	0.8 (0.2–2.9)
Dithiocarbamate	2.1 (1.0–4.3) ^a	2.5 (0.8–7.8)	1.6 (0.9–2.9)	2.2 (0.8–5.7)
Triazole	0.8 (0.4–1.7)	0.8 (0.2–3.5)	0.7 (0.4–1.5)	0.7 (0.2–2.5)
Copper	1.8 (0.8–3.7)	3.0 (0.8–11.7)	1.5 (0.8–2.8)	2.0 (0.7–6.1)
Sulfur	1.4 (0.7–3.0)	2.4 (0.7–8.6)	1.3 (0.7–2.5)	1.7 (0.6–4.8)
Herbicides				
Amide	1.5 (0.7–3.3)	1.6 (0.5–5.4)	1.5 (0.7–3.1)	1.4 (0.5–4.3)
Aryloxyphenoxypropionate	1.7 (0.5–5.5)	1.1 (0.1–12.9)	2.0 (0.7–5.8)	1.1 (0.1–14.8)
Benzoic acid	0.5 (0.1–1.9)	0.2 (0.1–2.1)	0.5 (0.1–1.7)	0.3 (0.1–2.5)
Carbamate	0.8 (0.3–2.2)	0.5 (0.1–2.4)	0.7 (0.3–2.1)	0.4 (0.1–1.8)
Dinitroaniline	0.6 (0.2–1.4)	0.5 (0.1–3.7)	0.7 (0.3–1.6)	0.7 (0.2–3.7)
Dinitrophenol	1.1 (0.6–2.3)	1.9 (0.7–5.0)	1.3 (0.8–2.3)	1.9 (0.9–4.1)
Nitrile	2.7 (0.9–7.5)	11.4 (1.4–93.7) ^a	2.2 (0.9–5.6)	5.8 (1.1–29.0) ^a
Organophosphorus	1.5 (0.8–2.7)	2.4 (0.9–6.0)	1.5 (0.8–2.5)	2.0 (0.8–4.6)
Phenoxy	1.8 (0.9–3.3)	2.9 (1.1–7.3) ^a	1.7 (0.9–3.0)	2.3 (0.9–5.3)
Picolinic acid	1.7 (0.8–3.7)	3.0 (0.7–11.8)	1.5 (0.8–3.1)	2.3 (0.8–7.0)
Quaternary ammonium	1.1 (0.6–2.0)	1.4 (0.6–3.1)	1.2 (0.7–2.1)	1.6 (0.7–3.4)
Triazine	1.1 (0.6–1.9)	1.9 (0.8–4.3)	1.2 (0.7–2.1)	2.1 (0.9–4.7)
Triazole	0.8 (0.3–2.1)	1.0 (0.2–4.9)	0.9 (0.4–2.1)	1.2 (0.3–5.0)
Uracil	1.2 (0.5–3.0)	0.2 (0.1–2.2)	1.2 (0.5–2.8)	0.3 (0.1–2.9)
Urea	1.1 (0.6–2.1)	2.4 (0.9–6.8)	1.2 (0.7–2.2)	2.0 (0.8–5.0)
Sodium chlorate	2.5 (0.9–7.6)	2.2 (0.6–8.2)	2.9 (1.0–8.2) ^a	2.7 (0.8–9.3)

ORs are based on separate conditional logistic regression models for matched sets comparing subjects exposed to each pesticide family to subjects who did not report that pesticide use (reference group); corresponding exposure frequencies in cases and controls are shown in Appendix 3. Subjects exposed to any pesticide for gardening exclusively are included in the analysis, but are treated as their own exposure group. All analyses are adjusted for Mini Mental State Examination and cigarette smoking (in pack-years). Arsenic, copper, sulfur, and sodium chlorate are inorganic compounds.

^a $p < 0.05$.

PD = Parkinson disease; OR = odds ratio; CI = confidence interval; AO = age at onset.

organochlorine insecticides and some herbicides (nitriles, triazines, ureas), but not phenoxy herbicides.

Analyses implementing MI in men overall and in men with older onset PD are presented in Table 3 and Appendix 4. In men, significant ORs were found for organochlorine insecticides and sodium chlorate, without significant dose-effect relations. No associations with younger onset PD were detected (data not shown). In men with older onset PD, significant associations were found for organochlorine insecticides, ni-

trile herbicides, and arsenic; significant trends were observed for organochlorine insecticides and arsenic, but not for nitrile herbicides. In multivariate analyses, both in men overall and in men with older onset PD, only organochlorines remained associated with PD after adjusting for other pesticide families. When the relation between professional pesticide exposure was adjusted for organochlorines among men with older onset PD, the OR for professional pesticide use became 2.3 (95% CI = 0.4–12.7).

Figure 2 shows the results of dose-effect analyses for organochlorine insecticides among men with older onset PD, using a complete-case analysis and multiple imputation; as shown in the figure, analyses using cumulative lifetime number of hours or years of exposure yielded similar results.

Figure 3 shows the relation between PD and organochlorines among men according to age at onset and period of exposure. Cases with older onset PD used organochlorines more frequently than matched controls at each time period, whereas there was no difference between cases with younger onset and their matched controls. In addition, there was no large difference in exposure frequencies between controls matched to each group of cases. Similar findings were observed for insecticides or professional pesticide use overall (data not shown).

Analyses stratified by median disease duration, analyses restricted to subjects with MMSE above the median, and lagged analyses excluding all exposures in the 5 years preceding onset in cases and the index date in controls led to similar results (not shown).

Discussion

As part of a community-based case-control study of PD among subjects characterized by a high frequency of professional pesticide exposure, using a detailed case-by-case expert exposure assessment, we found an association between PD and professional pesticide use with a dose-effect relation. In men, insecticides were more strongly associated with PD than fungicides and herbicides; the most robust association was found for organochlorines. Most of these associations were stronger in older onset PD and were characterized by dose-effect relations.

Previous studies suggested that pesticide exposure is associated with PD.^{7,8} A meta-analysis of case-control studies found an OR of 1.9.²⁰ Three cohort studies reported positive associations^{21–23}; in the largest, several chemical exposures were studied, but only pesticides were associated with PD (risk ratio = 1.7).²¹ In the Agricultural Health Study (AHS), incident ($n = 78$) but not prevalent ($n = 83$) self-reported PD was associated with pesticide use.¹⁶ Specific pesticides were associated with PD; among insecticides, ORs were increased for phorate (organophosphorus) and lindane

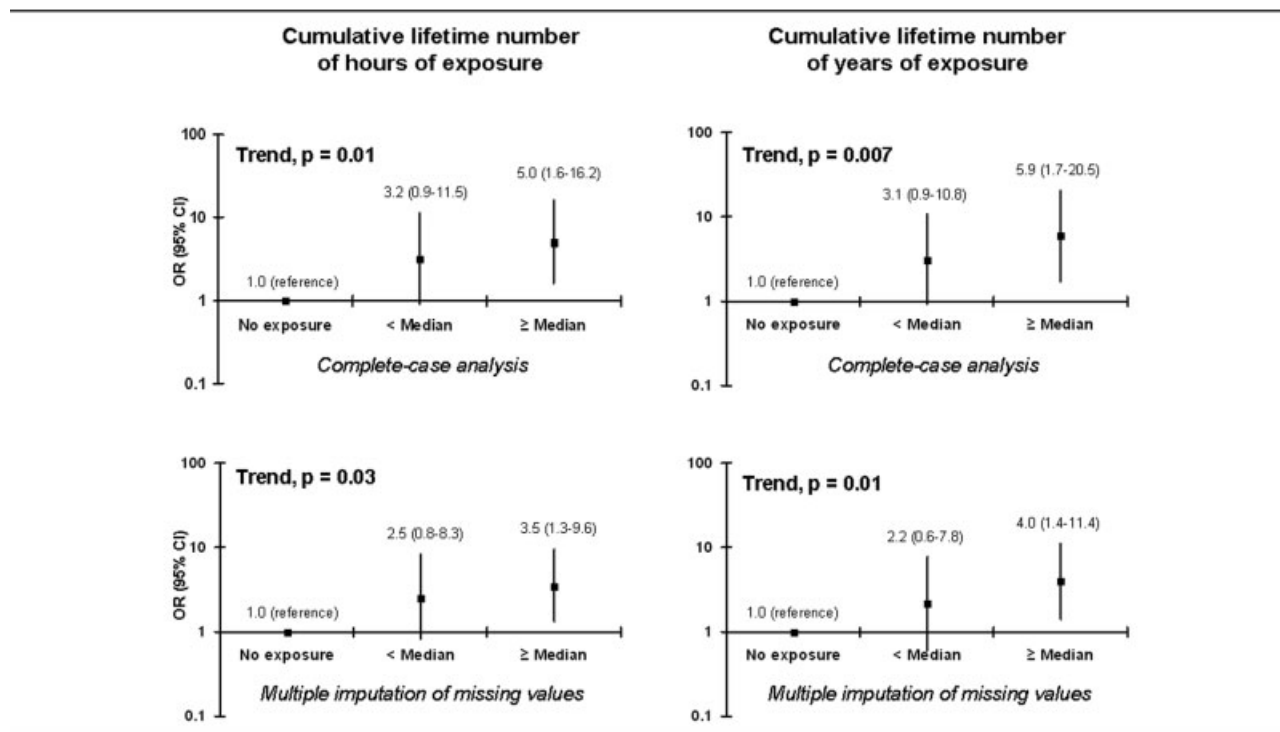


Fig 2. Association between Parkinson disease (PD) and exposure to organochlorine insecticides among men with onset of PD at age >65 years and matched controls: dose-effect analysis using a complete-case analysis and multiple imputation of missing values. Odds ratios (OR, 95% confidence intervals [CI]) were computed using conditional logistic regression models for matched sets. The reference category included subjects not exposed to organochlorine insecticides; subjects exposed to any pesticide for gardening exclusively are included in the analysis, but are treated as their own exposure group and excluded from the reference group. Analyses are adjusted for Mini Mental State Examination and cigarette smoking (in pack-years). The median of the distribution of the cumulative lifetime number of hours of exposure to organochlorines was 70 (IQ [interquartile] range = 159) in exposed cases and controls combined; the median of the distribution of the cumulative lifetime number of years of exposure to organochlorines was 16 (IQ range = 26) in exposed cases and controls combined.

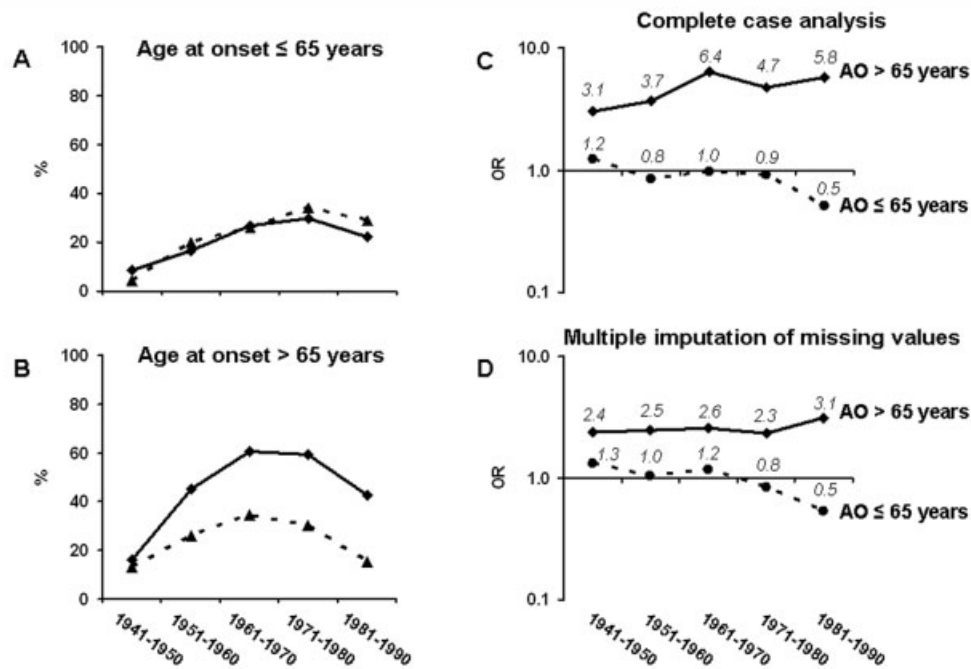


Fig 3. Association between Parkinson disease and exposure to organochlorine insecticides among men by period of exposure and according to age at onset (AO). Panels A and B show the frequency of exposure to organochlorines by time period in male cases with AO below and above 65 years and in their matched controls. (—) cases, (---) controls. Panels C and D show the corresponding odds ratios computed using conditional logistic regression models for matched sets, based on a complete case analysis or multiple imputation of missing variables. The reference category is represented by subjects not exposed to organochlorine insecticides; subjects exposed to any pesticide for gardening exclusively are included in the analysis, but are treated as their own exposure group and excluded from the reference group. Analyses are adjusted for Mini Mental State Examination and cigarette smoking (in pack-years).

(organochlorine; OR = 1.4 [0.8–2.5] for both). With the exception of the AHS,¹⁶ most studies included limited exposure assessment and few exposed subjects, precluding detailed studies of dose-effect relations and specific pesticides, and did not distinguish men from women.

The association between PD and professional pesticide use was stronger for older males. Using different age cutoffs to define this group led to the same conclusions. This finding is consistent with the view that genetic susceptibility plays a stronger role in younger onset cases, whereas environmental factors play a stronger role for older onset cases.²⁴ Alternative explanations can be considered. First, cases with older onset may have been exposed for more years than those with a younger onset; however, the number of years of exposure was similar between both groups for specific families of pesticides, and the cumulative lifetime hours of exposure were generally greater for younger onset cases; in addition, ORs were not significantly increased in younger cases with the highest exposures. Higher exposure levels in younger onset cases are due to increasing use in more recent decades; the size of the farms of the younger cases was also larger. Second, there are time-trends in exposure frequencies of pesticide families; these were higher in controls matched to older cases

than in those matched to younger cases for products such as arsenic, copper, or sulfur, which were used before more sophisticated products became available; however, they also had lower exposure levels for these products. Alternatively, controls matched to older cases had lower exposure frequencies for the majority of other families, including organochlorines. In addition, when studying the association between PD and professional pesticide, insecticide, or organochlorine use among men, we found that associations were present in men with older onset PD for all time periods, whereas there was no association for younger onset cases for all periods. Differences in exposure rates between controls matched to younger and older onset male cases did not explain the observed differences in ORs. Third, younger subjects may have been more educated with respect to pesticides than older subjects and may have taken more precautions while spraying.

We did not find an association between PD and paraquat (quaternary ammonium herbicide). In the AHS, paraquat was associated with prevalent but not incident PD.¹⁶ One study reported a significant association based on 4 exposed cases and no exposed controls²⁵; there was no association in 3 other studies.^{26–28} A Taiwanese study found an association between PD and paraquat use for >20 years.²⁹ There are several

possible reasons for these conflicting findings. First, in France, paraquat is mainly used as a nonselective herbicide to kill weeds around the fields, thus resulting in lower exposure levels compared to other herbicides. Second, if gene-environment interactions are involved, paraquat may be associated with PD only among susceptible individuals.^{9,30} Third, toxicological studies have suggested that maneb (dithiocarbamate fungicide) and paraquat act synergistically.³¹ We found an association between PD and dithiocarbamate fungicides among men, which was not significant in multivariate analyses or after multiple imputation of missing values; interaction analyses showed no synergy between both families (data not shown).

Among men, the most robust association with PD was for organochlorine insecticides. A German study reported an association between PD and organochlorines based on 9 exposed subjects.³² In a family-based study, organochlorines and organophosphates were associated with PD.³³ In Greenland, PD cases had higher levels of plasma dichlorodiphenyldichloroethylene than controls.³⁴ Postmortem studies showed that PD brains had higher dieldrin or lindane levels than control brains.^{35–37} These findings do not imply causality; they may simply reflect past pesticide exposure, because organochlorines are the only pesticides for which long-term biomarkers are available. Nevertheless, there is laboratory evidence in favor of dopaminergic neurotoxicity of organochlorines. Dieldrin induces apoptotic cell death, alters dopamine levels, and induces mitochondrial dysfunction and protein aggregation.³⁸ Endosulfan inhibits proteasomal activity.³⁹ Oral administration of lindane decreases dopamine levels in rats.⁴⁰ Lindane and dieldrin induce reactive oxygen species production in cultured microglial cells.⁴¹ In our study, the main organochlorine used was lindane, followed by dichlorodiphenyltrichloroethane (DDT). Lindane is the most common isomer of hexachlorocyclohexane and was used as a large spectrum insecticide in agriculture in France between the early 1950s and 1998; in our study, its main target was corn, followed by potatoes. It is characterized by a high persistency in the environment, with the potential for long-range transport. Significant levels of lindane and other organochlorines are still detected in the sera of humans.⁴²

A strength of our study is the use of a detailed exposure assessment method. It is however possible that farmers did not declare some products (eg, rarely used ones). Because cognitive impairment is more frequent in PD, recall may be differential. We therefore adjusted all our analyses for MMSE and performed sensitivity analyses restricted to subjects with higher MMSE. Alternatively, some cases may have been aware of a possible link between PD and pesticides, which may affect the exposure assessment overall; however, for specific pesticides, it is more likely that recall errors were nondifferential,

given the many pesticides reported, the lack of strong candidates, and the complex pattern of use.¹⁴

Most of the participants knew what crops the products they declared were used for and whether they were insecticides, fungicides, or herbicides; therefore, the proportion of missing values was low for this broad classification. Farmers were not always able to recall specific pesticides, and our exposure assessment method resulted in a number of missing values for pesticide families. The proportion of missing values was variable across families, and depended on disease status, MMSE, time period, age, and gender. To address this issue, we used MI to impute missing values based on these covariates. This approach is considered to decrease bias without strongly affecting standard errors of regression parameters.⁴³

We did not ask the participants to provide information on protective equipment use. This information is difficult to obtain, with farmers being likely to over-report their use when interviewed by occupational physicians. In addition, protective equipment was seldom used in France when the majority of the participants included in this study, even those with younger onset, started working; it is therefore unlikely that there have been large differences in use of protective equipment among study participants because of their period of activity and the type of farms they ran. It is only more recently that farmers trained in agricultural schools have started more systematically using protection when handling pesticides.

Accounting for exposure to multiple pesticides is a challenge for the analysis. We defined a priori pesticide families using a traditional chemical classification. This approach defines groups of subjects exposed to pesticides sharing chemical characteristics, and leads to groups of larger size than if we had considered pesticides separately; to further reduce the number of exposure variables, we restricted our analyses to families with at least 5% of controls exposed. However, the relation with the disease may be heterogeneous within a family, and this approach may suffer from a dilution of effects. Multivariate analyses identified organochlorines as the main variable associated with PD. This family had the highest frequency among all variables included in the model; therefore, it is possible that the lack of association with other less frequently used families may be due to insufficient power of the study.

To perform dose-effect analyses for pesticide families, we used the cumulative lifetime hours of exposure. In comparison to variables defined by the number of years or days of exposure, this variable captures information about duration of spraying (which is related to land size or type of spraying). Nevertheless, using number of years or days of exposure led to similar findings. Because owners of small farms may be more likely to spray pesticides using portable devices or small tractors compared to

owners of larger farms, we performed analyses using hectares-years, days, or hours, with similar results.

The timing of exposure may be important, but it remains challenging to define the appropriate exposure window, because it has been suggested that even developmental exposures may be relevant in PD.³¹ To exclude exposures that may have occurred during the pre-symptomatic phase, we performed lagged analyses, and found results similar to the overall ones.

In this study, we included cases with a recent diagnosis of PD (median disease duration since diagnosis of 1.5 years). We also performed analyses stratified by median disease duration, and found that association measures were similar in cases with shorter and longer disease duration; these findings are not in favor of a prevalence-incidence bias.⁴⁴

In conclusion, professional pesticide use was associated with PD in a population characterized by a high prevalence of exposure. These findings may provide leads for further toxicological studies. Our findings for organochlorine insecticides may be particularly important, because they are highly persistent in the environment; however, they exclusively concern professional pesticide exposure, and no conclusion can be drawn regarding other sources of exposure.

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